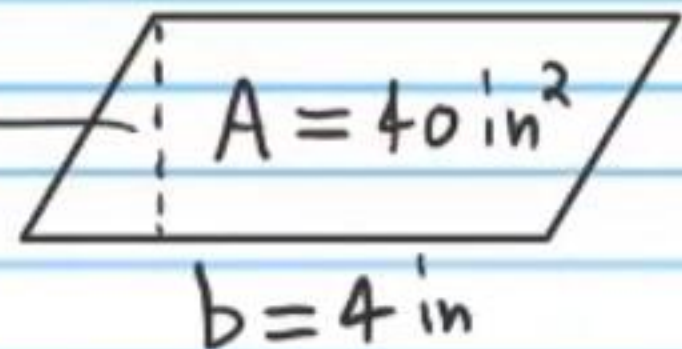


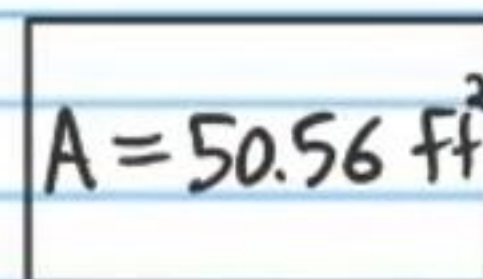
Applications of Algebra (Part II): Solving Equations (Homework)

① If the area of a rectangle is 63 cm^2 and the length is 7 cm , what is the width?

9 cm

② $h = ?$  $A = 40 \text{ in}^2$
 $b = 4 \text{ in}$

10 in

③  $A = 50.56 \text{ ft}^2$ $L = 6.4 \text{ ft}$
 $W = ?$

7.9 ft

④ Use the rate-distance-time formula ($r = d/t$) and the given info to find the missing value.
 $r = 45$, $t = 10$, $d = ?$

$d = 450$

⑤ How far would an object move if it were traveling at $31 \text{ miles per hour}$ for 1.4 hours ?

43.4 mi

⑥ 39% of a number is 0.0312 . What is the number?

0.08

⑦ 28 is 63% of what number?

44.4

⑧ Eleven is what percent of five?

220%

⑨ 6.44 ounces of a 7 ounce watermelon are water. What percent of this watermelon is water?

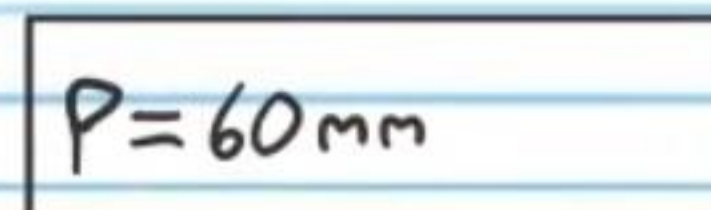
$92\% \text{ water}$

⑩ Andrew used 0.5% percent of his land to grow strawberries. If he used 45.115 acres of land to grow strawberries, how much land does he own in total?

$9,023 \text{ acres}$

⑪ What percent of 0.4 is 0.96 ?

240%

⑫  $P = 60 \text{ mm}$ $W = 18 \text{ mm}$
 $L = ?$

$L = 12 \text{ mm}$

⑬ If the perimeter and length of a rectangle are 7.8 centimeters and 3.2 centimeters respectively, what is the width of the rectangle?

$W = 0.7 \text{ cm}$

⑭ If a number is decreased by 95% and the result is 45 , what is the original number?

900

⑮ If a number is increased by 8% and the result is 51 , what is the original number?

47.2

⑩ If 5 is increased to 15, what is the percent increase?

150%

⑪ If 75 is decreased to 50, what is the percent decrease?

33.3%

⑫ After three years, Andy's pet snake grew from 16 inches to 25 inches. By what percent did the snake's length grow during this time?

56.25%

⑬ After Mr. Johnson's stock value decreased by seven percent, the net value was \$7,497.66. What was the value of his stock before the decrease?

\$8062

⑭ You loan your brother Bob \$286 so that he can buy a used suit. If you charge him 3% simple interest daily, he accumulates \$328.90 of debt. How much time passed since he borrowed the money?

5 days

⑮ You put \$15,238 into a savings account that pays simple interest yearly. After three years, the account grows to \$17,523.70. Calculate the interest rate for the account.

5%

⑯ Larry has \$239.54 in his bank account. He started with \$203 and made simple interest

paid yearly for nine years. What was the interest rate for the account?

2%

⑰ Determine the initial velocity of an object that starts at $x_0 = 3$, ends at $x = 5$, and moves for $t = 1$ seconds with an acceleration of $a = 10$. Use the following formula: $x = x_0 + v_0 t + at^2/2$.

$v_0 = -3$

⑱ If the sum of three consecutive even integers is 294, find the integers.

96, 98, 100

⑲ If the sum of four consecutive odd integers is 112, find the integers.

25, 27, 29, 31

⑳ The sum of five consecutive even integers is 340. Find these integers.

64, 66, 68, 70, 72